



Influence of a fishing harbour on coastal geomorphology of the southwest coast of India and predictions of its future trends

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ABSTRACT

In the present study, we estimated the shoreline change along Ponnani Fishing Harbour (PFH), the coastal part of Malappuram, India, using the Digital Shoreline Analysis System (DSAS) version 5.0. We also assessed the land use and land cover (LULC) changes in the gateway of Ponnani Fishing Harbour (PFH) and its surrounds using Geographical Information System (GIS) and Remote Sensing (RS) techniques. LULC estimated that vegetation cover was more in 2004 than in 2020, supported by the values shown by VSB (Vegetation with Scattered Built-up) and BSV (Built-up with Scattered Vegetation) show a significant areal change in the study area. On the comparison of erosion and accretion in the two zones of the study area, erosional activities are dominant in Ponnani, which is in the southern part. Based on the Linear Regression Rate (LRR) values, the maximum erosion rate was -7.41 m/year at Ponnani. The southern sector is also undergoing a maximum accretion rate of 8.11 m/year in the Ponnani Harbour region. The End Point Rate (EPR) estimated similar trends of the highest erosion -43.24 m/year and highest accretion, 37.34 m/year rates in the southern sector. The Net Shoreline Movement (NSM) values demonstrate that the shoreline retreated by -285.9 m, and the shoreline accreted by 132.32 m at the Ponnani Harbour area over the last three decades. The Kalman filter model predicted that the shoreline at Ponnani and Puthuponnani beaches might experience further erosion of approximately a 100-metre shift by 2030. An increase in the erosion rate could erode another 140 m from the shoreline by 2040. The Ponnani and Puthuponnani beach extents must be considered high erosion risk zones in the perspective of the estimated results. This assessment indicates that the southern stretch of the area under the study of the PFH coastline is eroding. The significant changes in land use and land cover that have been seen over the past few years following the construction of PFH are expected to be the main cause of the severe loss of the coast in the southern part. The main reason for erosion in PFH would be construction operations that put pressure on the coastline side. The limited supply of sand used to replenish the coast naturally causes shoreline change due to the large drop in sediment load from the river, which is evident by the shrinking of the braided island size. The methods employed in this study and the results might help manage the PFH coast sustainably. The outcomes of this assessment suggest that similar studies should be conducted in other regions of India's western coastline to understand the shoreline evolution and develop a better coastal management strategy.

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1. Introduction

Land use denotes the anthropological activity or economic function related to a substantial portion of land, whereas land cover

refers to the feature existing on the earth's surface (Lillesand et al., 2015; Arunkumar and Joseph, 2015). Both land use and land cover are closely interrelated and are transposable. Understanding land use/land cover can be directly used for the progress of an area through land-use planning (Sahai et al., 1985; Arunkumar and Joseph, 2015).

Shorelines are the land and sea border, subject to continuous transformation due to their active environmental setting

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(Mentaschi et al., 2018; Baig et al., 2020; Natesan et al., 2015). It is one of the most significant active areas globally because it is susceptible to erosion, floods, storms, and hurricanes (Garcia-Rubio et al., 2012). Erosion makes coastal areas more vulnerable, which can be hazardous for activities carried out by people along the coast. Additionally, the increasing frequency of coastal disasters makes the beaches extremely sensitive and unpredictable (Saxena et al., 2013). The loss of land area due to the regressive movement of sand towards the ocean due to natural forces like wind, tides, waves, ocean currents, etc., changes the shoreline (Sheeja and Vishnu, 2015). Shoreline computation is one of the most critical constraints in understanding coastal erosion and deposition and investigating coastal morphodynamics (Armenio et al., 2019).

The surface temperatures of nearby bodies of water can occasionally rise or fall due to coastal development and anthropogenic activities like logging, agriculture, construction, mining, drilling, dredging, and tourism. Development can produce sediment that reduces streamflow and direct deposition of sediments, causing damage attributed to global warming (Bast, 2010). Removing material from the shoreline by wave action, tidal currents, and human activity is known as coastal erosion, and it often results in a coastline that is moving inland (Mandal, 2018). A coastline advances where the deposition of sediment exceeds the rate of erosion, or where there is a rise due to the uplift of the land or a fall in sea level, and retreats as the result of erosion exceeding deposition, or where there is submergence due to land subsidence or a sea-level rise (Bird, 2011). Coastline changes can be measured over various timescales, ranging from the past few thousand years down through recent centuries or decades to the annual or seasonal fluctuations and short-term modifications related to the different tidal cycles caused by particular weather events (Bird, 2011). Besides this, coastline changes can be determined for various timelines using RS (Remote Sensing) and GIS (Geographic Information System) technologies. Tidal activity, landslides, and natural changes in coastal areas trigger coastal change. It adversely affects the social and economic life of the human race and its vitals. The present-day surface of the earth is the consequence of geological processes over a long period. The current state of the beaches is the result of long-term changes from the beginning of the ice age, which began about 13,000 years ago. The sea level at that time must have been about 137 m below today's. As the ice melted, it rose at about one metre per century (Milliman and Emery, 1968).

India's coastal areas are home to around 611 million people (www.indiaonlinepages.com, census report of 2018). Coastal hazards, particularly coastal erosion, are a major concern in many densely populated areas of the country. According to a study based on satellite-derived Shoreline Change rate (Mohanty et al., 2017), West Bengal has the greatest percentage of erosion (70%), followed by Kerala (65%), Gujarat (60%), and Odisha (50%). Kerala is situated at the southwestern tip of the Indian peninsula with a coastline of 560 km along the eastern margin of the Arabian sea. The Kerala coast is a narrow stretch in the south-western part of peninsular India and is highly dynamic in nature (Thankappan et al., 2018; Perabi et al., 2022; Chowdhury and Behera, 2017; Nair et al., 2015; Pradeep et al., 2022; Kadavan and Joseph, 2018; Samsuddin and Suchindan, 1987). Several studies reported long-term shoreline changes as well as coastal environmental signatures through this stretch. The Kerala coast is highly vulnerable to shoreline changes and coastal hazardous calamities such as cyclones, floods, and tsunamis. The shoreline of the Thiruvananthapuram district is the most prone to erosion amongst the nine coastal districts: Kasaragod, Kannur, Kozhikode, Malappuram, Thrissur, Alappuzha, Ernakulam, Kollam, and Thiruvananthapuram (Deepchand, 2020). The current study region is part of the Malappuram district, which is also part of the hazardous coastal district of the state. Because

Malappuram is the state's most populated district, hazardous activities in densely populated areas such as coastal zones are also significant and vital to research to develop appropriate rehabilitation measures. Using GIS tools, Paravath et al. (2021) investigated the effects on coastal geomorphology along the Ponnani coast. The results of the investigations indicate that the breakwater's south and north end experienced considerable effects both before and after construction.

With the convenience of high-resolution satellite remote sensing data, the studies of coastal land use/land cover and shoreline changes have converted easier, which have shown to be of extreme value in all fields of earth sciences due to the swift, tedious, synoptical, and multispectral coverage. Satellite images have stood out successfully for monitoring shoreline changes in different parts of India (Solán et al., 2001; Arunkumar and Joseph, 2015).

The study uses GIS and RS techniques to assess the land use and land cover changes in PFH and its surroundings. The present study investigated the shoreline change along the PFH coastline Digital Shoreline Analysis System (DSAS) version 5.0, extension to ESRI Arc GIS v. 10.8 (Thieler et al., 2009). The objectives of the study are (a) to delineate land-use and land-cover map of 2004 & 2020 from Google Earth in and around the PFH area. (b) To compare the land-use and land-cover changes, (c) to recognize the zone where substantial coastal erosion occurs, (d) enumerate the rates of shoreline changes along the study area, and (e) classify a coastal stretch based on shoreline change rates into erosion, accretion, and stability, (f) decipher major reasons behind the shoreline changes, (g) to predict future shoreline position using advanced mathematical tools, (h) suggest appropriate rehabilitation measures to overcome the hazardous situations caused by shoreline changes.

DSAS is an open-source software plugin that works within the Environmental Systems Research Institute (ESRI) GIS software (Arc GIS) which computes rate-of-change for a time series of shoreline vector data. The statistical parameters that compute the shoreline change, such as Shoreline Change Envelope (SCE), Net Shoreline Movement (NSM), End Point Rate (EPR), Linear Regression Rate (LRR), Weighted Linear Regression Rate (WLR), and Least Median of Squares (LMS) are the key functions developed in DSAS using statistical equations (Thieler et al., 2009). The present study uses Linear Regression Rate (LRR), End Point Rate (EPR), and Net Shoreline Movement (NSM) to calculate the shoreline change rate.

2. Study area

The shoreline belt of Ponnani Fishing harbor (PFH) was selected for land-use/ land-cover change and shoreline change analysis in Malappuram district, one of the nine coastal districts in Kerala (Fig. 1). PFH is located in Ponnani municipality in Ponnani taluk. It is situated near the river mouth and the estuary of Bharathapuzha, bounded by a series of brackish lagoons in the north and the Arabian Sea in the west. Along the Ponnani coastline, several beaches can be located. The Tidal Doorway of Bharathapuzha is located at Padinjarekkara Beach, where the Tirur River joins Bharathapuzha and drains into the Arabian Sea. On the southern side of the Bharathapuzha river mouth, the centre of the PFH is located. The geographical extent of the study area is 52.17 km² and lies between the north latitude of 10°45'00" to 10°50'00" and the east longitude of 75°52'30" to 75°57'30". The study area deceits in the equatorial region and is mainly exposed to the humid tropical wet climate. Ponnani has an average daily temperature of around between 31.2°C and 25.2°C.

April is the warmest month, with a maximum average temperature of 32°C. The total average annual precipitation is about 2852 mm. With an average of 4 mm of rainfall, January is the driest month of the year. The wettest month is July having an av-

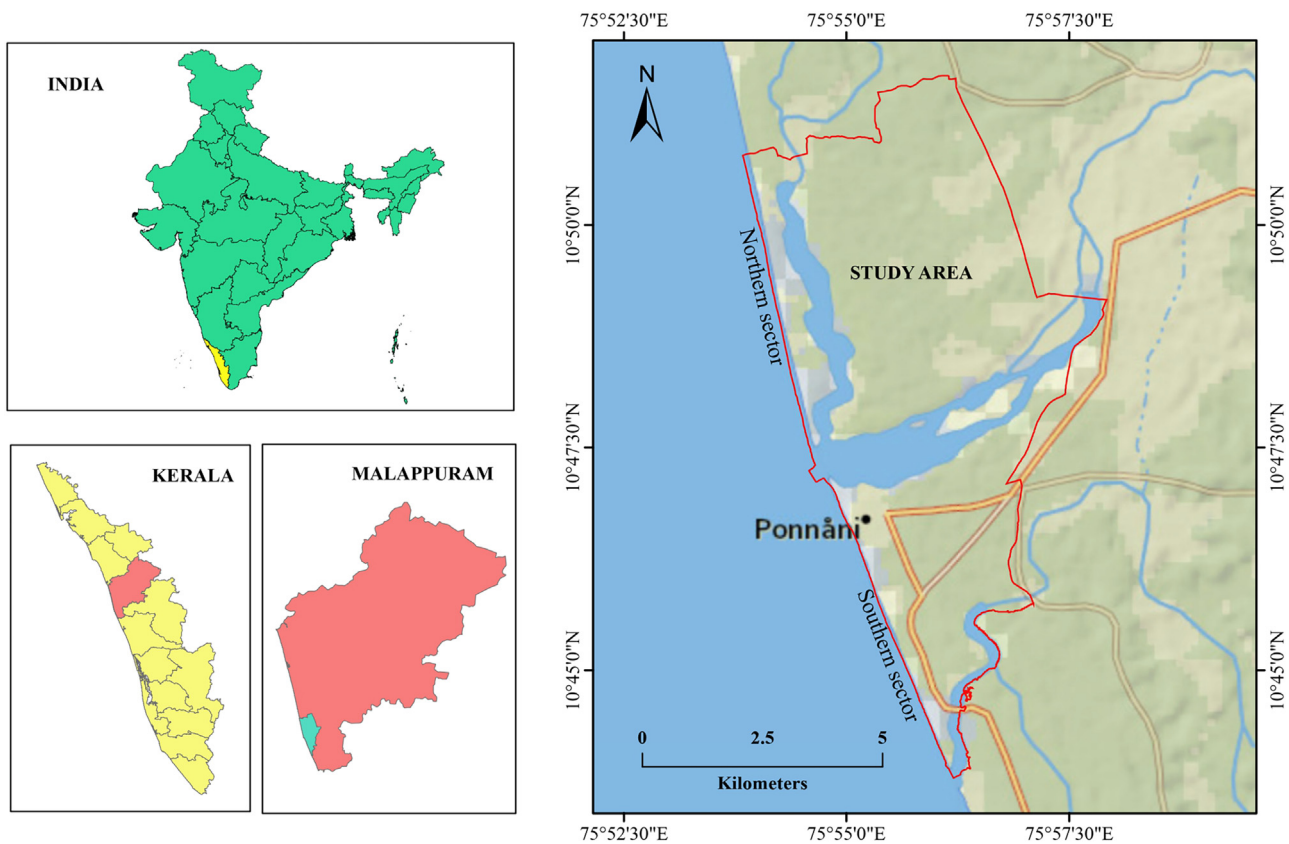


Fig. 1. Study area map.

erage of 702 mm of precipitation. The average annual weather in the study area is high, low, and moderate temperatures and average rainfall throughout the year. In the district, the South-West and North-East monsoons are responsible for about 120 mm of the yearly rainfall. The South-West Monsoon provides more rain to regions inland and on the coast. In the district, crystalline and tertiary formations predominate, with alluvial patches, while most of the shoreline is rocky and lateritic. However, the Ponnani section is made entirely of alluvium. The most commonly found soil is sand to sandy clay variety, confined mainly to coastal areas. Low-land soils, called alluvial soils, are found primarily in valleys and plains along the shore. The most potential aquifer in the district is the alluvial aquifer. The coastal alluvium is essentially composed of sand, silt, and clay. The groundwater in these aquifers occurs under water table conditions.

The warmest month is April, with an average maximum temperature of 32°C. The total average annual precipitation is about 2852 mm. The driest month of the year is January, with an average of 4 mm of rainfall. The wettest month is July having an average of 702 mm of precipitation. The average annual weather in the study area is high, low, and moderate temperatures and average rainfall throughout the year. The district receives about 1201 mm of rainfall annually from the South-West and North-East monsoons. The South-West Monsoon brings more rain to locations that are inland and on the coast. The major portion of the coast is mostly rocky and lateritic on crystalline and tertiary formations predominate in the district, with alluvial patches. Still, the Ponnani stretch is composed of pure alluvium. The most commonly found soil is sand to sandy clay variety, confined mainly to coastal areas. The most potential aquifer in the district is the alluvial aquifer. The coastal alluvium is essentially composed of sand, silt, and clay. The groundwater in these aquifers occurs under water table conditions.

Table 1

Data Sources used for the study.

Sl. No	Data source and sensor	Date of acquisition (DD/MM/YYYY)
1	Survey of India Toposheet (1:50,000 scale)	15/07/1987
2	Landsat 7 ETM+ C2 L2	14/03/2005
3	Landsat 7 ETM+ C2 L2	23/01/2010
4	Landsat 8 OLI C2 L2	20/04/2013
5	Landsat 8 OLI C2 L2	01/09/2015
6	Landsat 8 OLI C2 L2	28/11/2018
7	Landsat 8 OLI C2 L2	17/11/2020
8	Google earth images	2004 and 2020

3. Data sources

This study evaluated the changes in coastal erosion from 1987, 2005, 2010, 2013, 2015, 2018, and 2020 using integrated remote sensing, GIS, and DSAS technologies. For this study, the Survey of India Toposheet (1:50,000 scale) dated 15/07/1987, ETM+ (Enhanced Thematic Mapper Plus), and OLI (Operational Land Imager) datasets acquired from 2013 to 2020 with 30 m spatial resolution are used to delineate the shoreline changes along the PFH coast. Table 1 indicates the data sources used for shoreline change detection.

4. Methodology

Integrated GIS, RS, and field-based studies are used throughout the study to determine significant land use-land cover change and shoreline change in the study area over a reasonable time period. The study also investigated the key factors that cause LULC and shoreline changes, which seriously threaten the ecosystem and human settlement. Advanced mathematical models forecast future

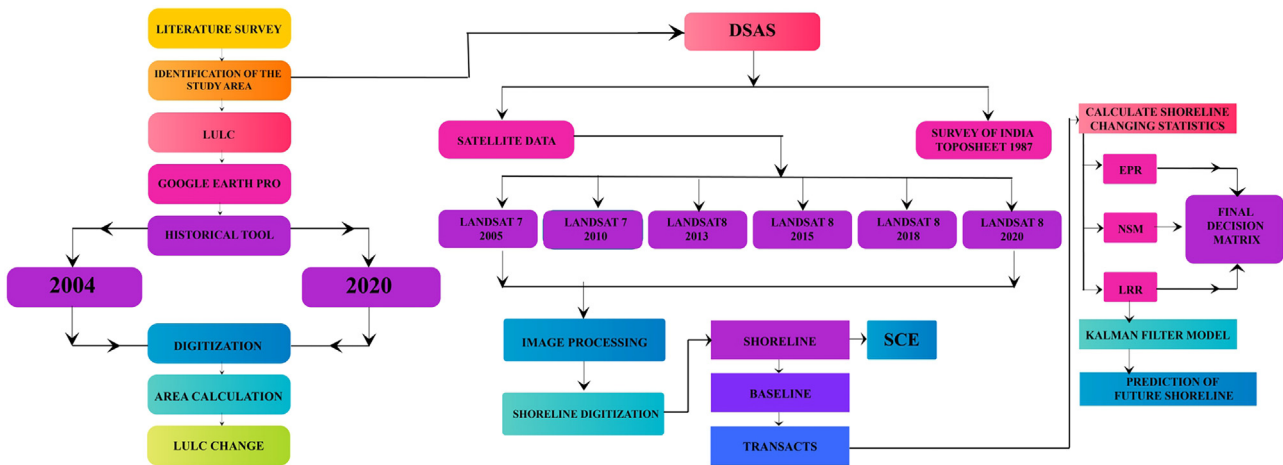


Fig. 2. Flow chart showing detailed methodology adopted for the study.

probabilistic shoreline positions and provide appropriate mitigation measures to protect the environment from further hazardous developments. Fig. 2 depicts the detailed methodology used in the investigation.

4.1. Land Use-Land Cover (LULC) change, 2004–2020

For analysing the temporal evolution of land use land cover in PFH and surrounding area for two different timelines, 2004 and 2020, the first thing to consider is the RS data to extract the information about the area of interest (AOI) for the two different timelines. The proposed methodology includes two main parts: (i) Remote Sensing data collection, such as Google Earth Images of AOI – 2004 and 2020 and other relevant materials, and (ii) data verification, preparation/ processing, re-projection/setting of the coordinate system, clipping, digitizing, classification, etc., and importing data to personal/file geo-databases.

4.2. Coastal erosion analysis

4.2.1. Shoreline change analysis

The Survey of India (SOI) toposheet (1:50,000 scale), ETM+(Enhanced Thematic Mapper Plus), and OLI (Operational Land Imager) images acquired from 1987 to 2020 were used as the primary data source for shoreline extraction. Considering differences in the altitude, attitude, and velocity of the sensor platform induce several defects in satellite images, such as radiometric distortion, geometric distortion, and the presence of noise, the images were subjected to pre-processing operations. To enhance the quality of the imagery, image correction methods such as geometric correction, cloud correction, and noise reduction were used. After pre-processing, the satellite images are geo-referenced and projected using polygonic projection with WGS 84 as the datum using ArcGIS 10.8 software. The discrete shorelines were extracted from the multi-temporal images, and their geo-database was prepared using ArcGIS 10.8 software. The movement of the shoreline concerning the position of the reference line (baseline) is considered a landward shift (erosion) and seaward shift (accretion) at each transect of the segments, and the statistical values of the measurement have been denoted as negative (-) for erosion rate and positive (+) for accretion rate (Anders and Byrnes 1991). After that, the different periods of shoreline data have been placed into DSAS for further computation of shoreline change for 1987, 2005, 2010, 2013, 2015, 2018 and 2020.

4.2.2. DSAS (Digital shoreline analysis system)

DSAS is an extension of ArcGIS 10 software developed by the United States Geological Survey (USGS) and the TPMC Environmental Services (Thieler et al., 2009). The pre-defined statistical algorithm of this software permits the user to measure the shoreline change rate using geo-rectified multiple time series shorelines in each transect at a user-defined interval. In the DSAS tool, shoreline positions are assembled with five attribute fields which comprise Object ID (a whole number assigned to each transects), date (original survey year), shape (polygon), shape length, ID, and uncertainty values. Shorelines extracted from the different times were combined as a single feature in the attribute table, which generated a single shapefile of the several shorelines (Baig et al., 2020). By carefully digitizing the direction and shape of the outer shoreline, the baseline is obtained, and the cross-shore transects are created for calculating the shoreline change. A total of 477 transects were made, and the data from the transect feature class were used to calculate the variations. After that, DSAS computes the total transects for the coast (Baig et al., 2020).

4.3. Calculating change statistics

The information can be used to compute change statistics when the transect feature class has been created and all changes or modifications have been made. Usually, the DSAS executes five statistical operations, namely Shoreline Change Envelop (SCE), Net Shoreline Movement (NSM), End Point Rate (EPR), Linear Regression Rate (LRR), and Weighted Linear Regression (WLR), to measure the shoreline change rate in long-term and short-term basis (Thieler et al., 2009; Baig et al., 2020). In this study, the shoreline change rate on the PFH coast has been discussed in SCE, NSM, EPR, and LRR statistical methods.

With a position roughly 50 m behind the shorelines, an on-shore baseline was established to assess the shoreline trend. Based on the baseline and shoreline, the erosion/accretion regimes were computed for 50 m orthogonal transects along the coastline. Finally, the obtained erosion and accretion rates for the PFH coast were divided into seven categories: Very High Erosion, High Erosion, Moderate Erosion, Stable Coast, Moderate Accretion, High Accretion, and Very High Accretion.

4.3.1. Shoreline change envelop (SCE)

The SCE values measure the long-term net shoreline change in meters (distance) between 1987, 2005, 2010, 2013, 2015, 2018, and 2020. The SCE is estimated as the distance between the shorelines at each transect that is the farthest and closest to the baseline.

The rate of erosion and accretion of the coast has been quantified in relation to baseline positions, and the overall changes in shoreline movement have been characterized for all available shoreline positions.

4.3.2. Net shoreline movement (NSM)

Net shoreline movement (NSM) is another statistical parameter that calculates the distance of shoreline movement between the oldest and youngest shorelines at the point of intersection in each transect (Himmelstoss et al., 2018). They are also used to measure the long-term net shoreline change. It can be calculated using Eq. (1)

$$NSM = \{d_{2020} - d_{1987}\}m \quad (1)$$

4.3.3. End point rate (EPR)

EPR is calculated by dividing the distance of shoreline movement by the time elapsed between the earliest and latest measurements (Bouchahma and Yan, 2014). The EPR is a method for estimating the per-year rate of shoreline change. It is calculated by dividing the difference between the first (closest) and last (farthest) shoreline to the baseline at a specific transect and dividing it by the total number of years. In DSAS analysis, the EPR statistic is performed by dividing the distance of net shoreline movement (NSM) at transects (xn) by the total period of interval elapsed between the oldest and most recent shorelines. The magnitude and cyclical trends of the EPR values represent erosion and accretion processes in the coastal zone. It is calculated using the following formula Eq. (2)

$$EPR = \frac{d_{2020} - d_{1987}}{t_{2020} - t_{1987}} \text{ m/yr.} \quad (2)$$

4.3.4. Linear regression rate (LRR)

The most common statistical method for expressing shoreline movement and estimating change rates is LRR. The LRR represents the trend of shoreline changes during the short-term periods. This is determined by fitting a least-squares regression line to all shoreline points for a particular transect (Himmelstoss et al., 2018). The slope of the regression line is placed according to the sum of the squared residuals calculated from the values of squaring the offset distance of each data point from the regression line and adding the squared residuals together (Salghuna and Bharathvaj, 2015). This method uses all the data, regardless of trend or precision changes.

4.4. Prediction of future shoreline position

The future approximate shoreline position of the study area in the years 2030 and 2040 was prepared using the Kalman filter model function in the DSAS plugin. The Kalman filter model is a statistical model which uses linear regression rate (LRR) to predict future shoreline position (Long and Plant, 2012; Natarajan et al., 2021). The technique combines historic and model-derived shoreline positions to forecast future shorelines with an uncertainty band. The Kalman filter analyses the difference between the modelled and observed shoreline position to improve the forecast by considering rate and uncertainty (Long and Plant, 2012). We used uncertainty bands to anticipate the coastline positions of the study area between 2030 and 2040.

5. Results and discussion

The land use land cover map of PFH and adjacent area for 2004 and 2020 was prepared by digitizing the satellite images from Google Earth Pro-software using the historical imagery tool. Eight

Table 2
LULC change in area 2004–2020.

Category	2004 (km ²)	2020 (km ²)
Channel Islands	4.15	1.69
Open land	0.48	0.82
Wetland	4.26	2.32
Breakwater	0.01	0.01
Beach	0.67	0.52
Waterbody	5.71	9.02
Vegetation with Scattered Build-ups	29.59	3.24
Build-ups with Scattered Vegetation	7.24	34.52

land use -land cover forms were delineated and digitized using create polygon tool. The area representing each polygon was calculated and compared to the land use land cover change in terms of area for 15 years of a time gap in the study area. Fig. 3 shows land use-land cover maps of the study area in 2004 and 2020 and a comprehensive bar diagram of land use land cover change over this period.

For shoreline monitoring investigation, we divided the study area into two sectors- viz northern sector and southern sector, i.e., from Mangalam and Purathur north and from Ponnani and Puthuponnani to south for better interpretation of the estimated rates of erosion and accretion between 1987 and 2020. The vector layers of shorelines were overlaid, and erosion/accretion rates were calculated at 50 m intervals along the PFH coast using the DSAS plugin. The result showed that most of the coastal tract in the southern sector is susceptible to erosion. It can be seen that erosion is predominant in Ponnani, the south area of the study region, by comparing erosion and deposition in the two zones. Based on the LRR values, the maximum erosion rate was −7.41 m/year at Ponnani. The maximum accretion rate was 8.11 m/year in the Ponnani Harbour region and the Sothern sector. The maximum erosion rates (−43.24 m/year) and accretion (37.34 m/year) in the southern sector were calculated by the EPR to have similar values. The NSM measurements show that over the past three decades, the shoreline of the Ponnani Harbour region has receded by −285.9 m and intruded by 132.32 m. Fig. 4 shows the position of shorelines in representative years in the northern and southern sectors for 1987 and 2020.

5.1. LULC change, 2004–2020

The land use-land cover characteristics of the PFH coast are given in Table 2 and Fig. 3. The coastal villages like Mangalam, Purathur, Ponnani, and Puthuponnani are thickly populated, leading to significant socio-economic problems. Land use features include roads, rivers, channel islands, open land, wetlands, breakwater, beach, waterbody, VSB (Vegetation with Scattered Built-up), and BSV (Built-up with Scattered Vegetation) has been digitized. Changes are noted in the LULC in the AOI. The total extent of the study area is 52.14 km². The majority area is covered by VSB (Vegetation with Scattered Built-up) of 29.59 km² (56.74%), and the lowest area by a breakwater of 0.016 km² (0.03%). The BSV (Built-up with Scattered Vegetation) covers about 7.24 km² (13.88%). It shows that vegetation cover was more in 2004 compared to 2020. In 2020, the BSV covered the majority of the study area at 34.52 km² (66.17%) and the least covered by the breakwater at 0.017 km² (0.03%). The area covered by the VSB is 3.24 km² (6.21%). In 2020, many of the channel islands seen in 2004 vanished. They are partially and fully drowned in water, seen through Google Earth. New roads were constructed, and wetlands decreased. The VSB and BSV show a significant areal change in the study area. The BSV for 2004 was 13.88%, which increased to 66.17% in 2020; likewise, the VSB decreased from 56.74% to 6.21%, indicating vegetation cover loss to built-up. There is a great loss of channel islands in 2020 com-

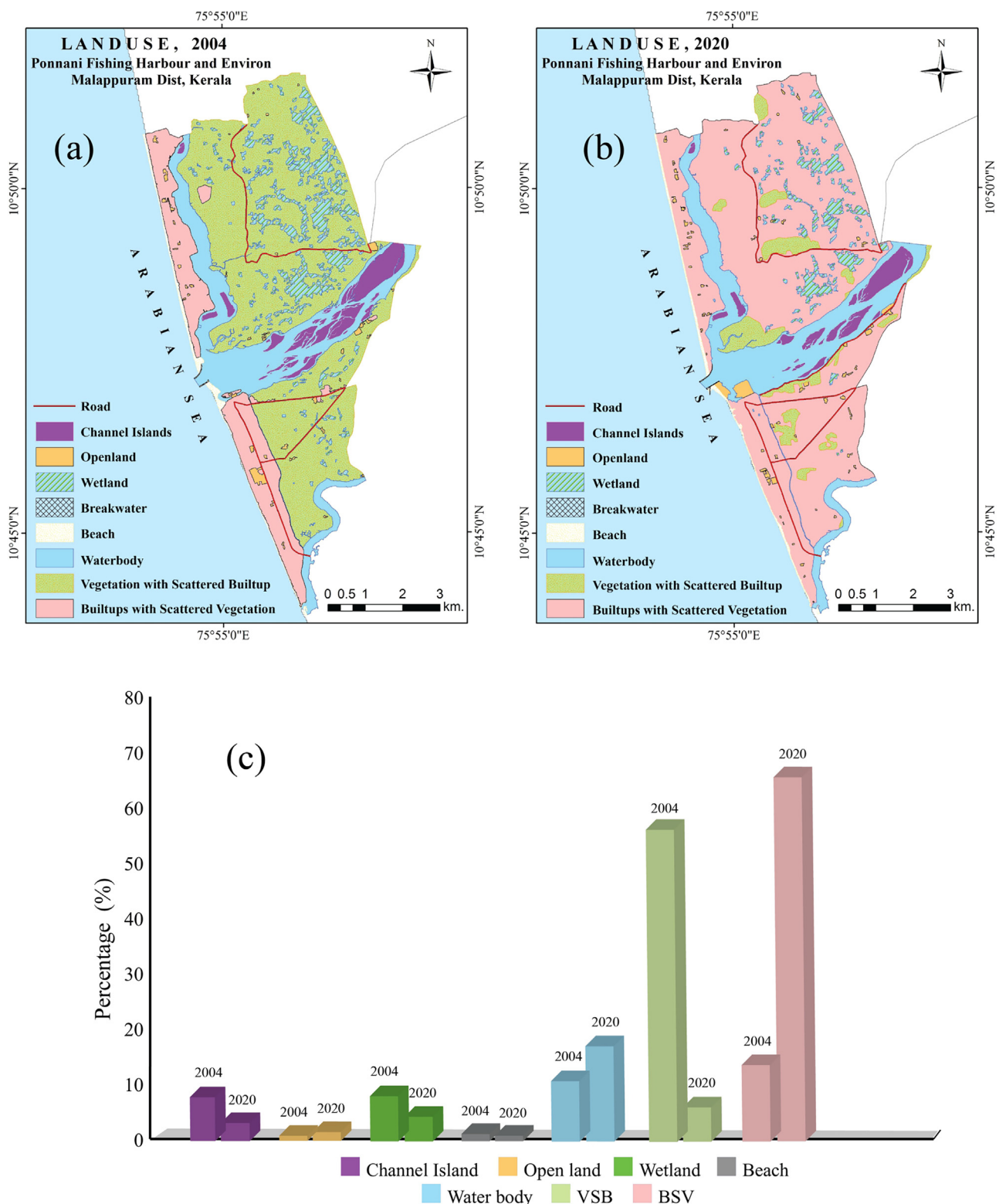


Fig. 3. (a, b) Land use/land cover maps of the study area in 2004 and 2020. (c) Bar diagram showing land use/land cover change in the given period of investigation.

pared to 2004. They vanished or partially drowned in water. Wetland showed a reduction in area. The waterbody area increased from 10.96% to 17.28% (2004 to 2020). An increase in beaches and shorefronts in the northern sector and a decrease in the southern sector showed that coastlines change due to natural and anthropogenic activity.

Small-scale industrialization, population dynamics, tourism, habitat facilities, coastal disasters, road widening, development of hospitals, external market forces, globalization of the economy, changes in culture and lifestyle, policies, and interventions are the most critical driving forces influencing the land use/land cover in the PFH. This coastal region is characterized by high population

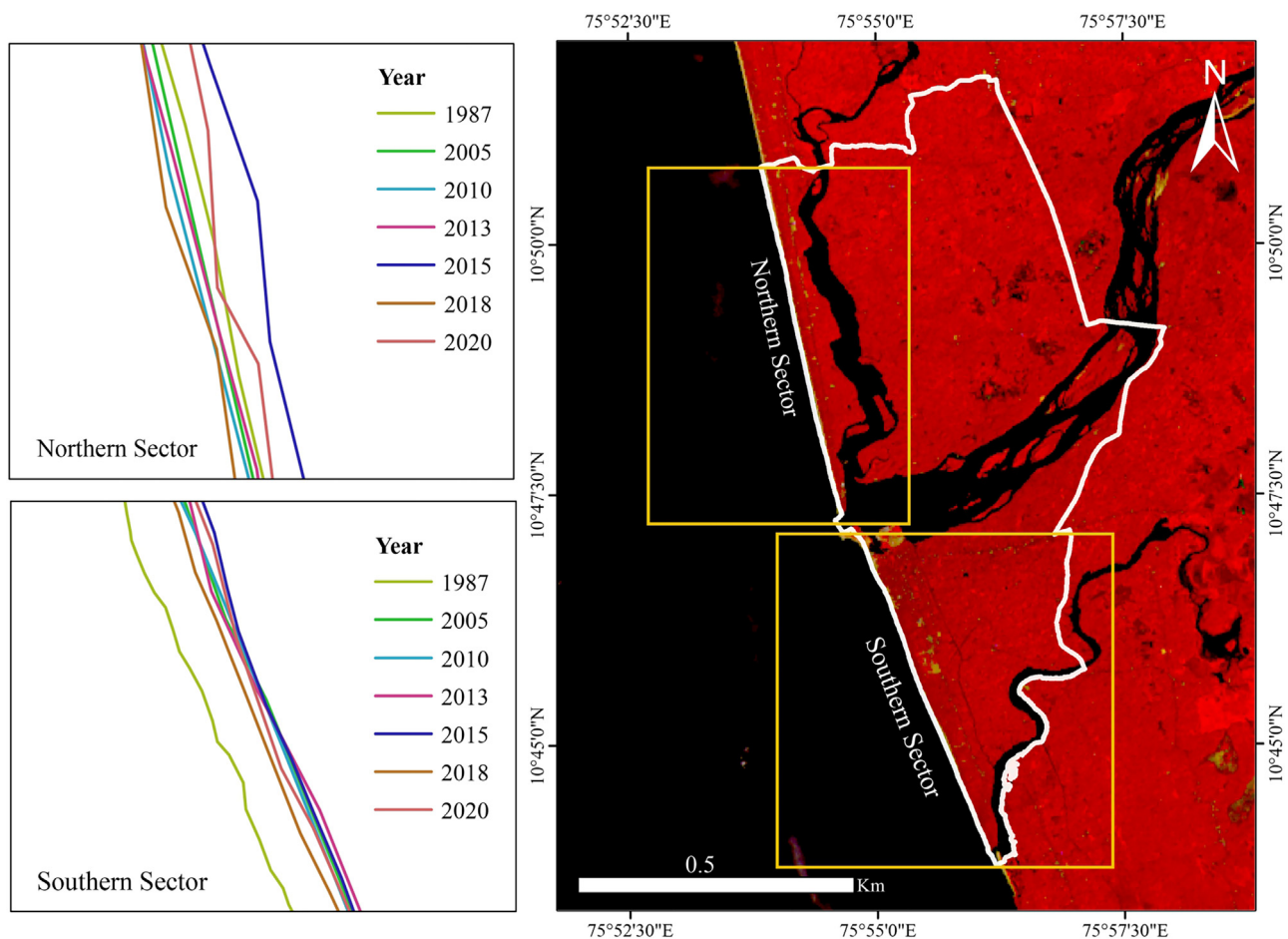


Fig. 4. Position of shoreline in representative years in northern and southern sectors of the study area.

density and hotspot areas for rapid and intense socio-economic and environmental changes, which require immediate policy responses. Thus, integrating information on existing land use/land cover is essential for planning. Data from land use/land cover studies can be used directly for land development through scientific planning (Sahai et al., 1985; Arunkumar and Joseph, 2015).

5.2. Shoreline change characteristics

5.2.1. Shoreline change envelope

The SCE values measure the long-term net shoreline change in meters (distance) between 1987, 2005, 2010, 2013, 2015, 2018, and 2020. The SCE is calculated as the distance between the shorelines at each transect farthest and closest to the baseline. SCE measures the shoreline change in distance, the greatest distance is 285.9 found at Ponnani harbour beach, and the minimum length is 20.5, which is in Mangalam, and the average distance value is 104.6.

5.2.2. Net shoreline movement

Net shoreline movement also calculates the distance of shoreline movement between the oldest and youngest shorelines at the point of intersection in each transect. They are also used to measure the long-term net shoreline change. In NSM, it measures negative and positive distances as erosion and accretion. The maximum negative distance is -285.9 m, and the positive distance is 132.32 m. The average negative distance is -87.87 m, and the average value of the positive distance is 24.60 m. Fig. 5 shows the dif-

ferent degrees of erosion and accretion in the study area between 1987 and 2020 using NSM with erosion and accretion profiles.

5.2.3. Rate of shoreline change by end point rate method

The EPR values show erosion and accretion in the study area, and most transect shows accretion. From the 477 transects, 408 transects are erosional, and 69 transects show accretion by the EPR method. According to the endpoint rate method, the figure shows the shoreline erosion and accretion in the PFH coast end-point. The erosional transects are found on Ponnani and Puthuponnai beaches. The maximum erosion rate is -43.24 and recorded in Ponnani harbour beach. The average erosion value of the coastal tract is -2.76 m. So, a total of 85.53% are erosional transects. In the accretion case, the maximum accretion rate is 37.34 , recorded at Ponnani fishing harbour. The average of all accretion rates is 1.60 , and the total percentage of accretion transects is 14.46%. High erosion is observed at the Ponnani coast near the harbour with a -43.24 erosion. This coastal tract is very prone to erosion in the future time. Fig. 6 shows the estimations of different degrees of erosion and accretion in the study area between 1987 and 2020 using EPR with erosion and accretion profiles.

5.2.4. Rate of shoreline change rate by linear regression method

The LRR represents the trend of shoreline changes during the short-term periods. This is determined by fitting a least-squares regression line to all shoreline points for a particular transect. From 477 transects, 348 are erosional and 129 accretional. The maximum value of erosion is -7.41 , and the maximum accretion value

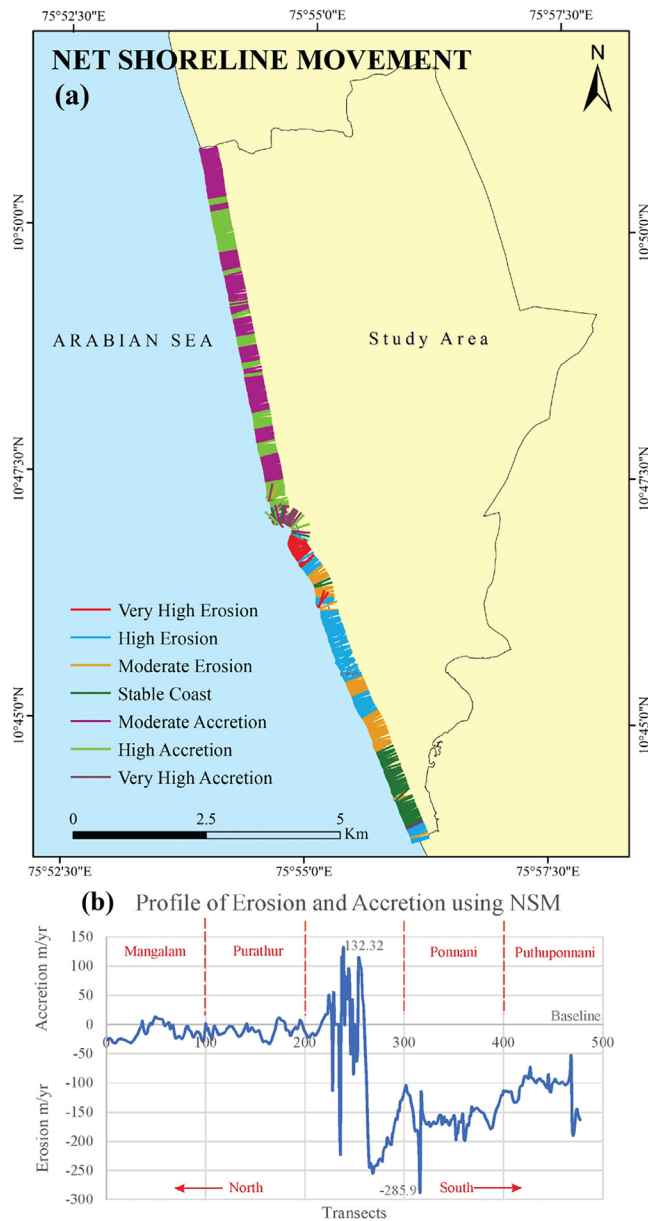


Fig. 5. Estimations of shoreline changes using NSM and profiles of erosion and accretion.

is 8.11. The percentage of all erosional transects is 72.95%, and the percentage of all accretion transects is 27.04%. The average of all erosional rates is -2.71 , and all accretion rates are 0.86 . Method. The most vulnerable erosion areas on this map are Ponnani and Puthuponnai. Fig. 7 shows the estimations of different degrees of erosion and accretion in the study area between 1987 and 2020 using EPR with erosion and accretion profiles.

5.2.5. Overview of shoreline changes in southern and northern sectors

5.2.5.1. Southern sector. This sector was dominated by erosion and did not observe any accretion. The greatest erosion rates in the Puthuponnani and Ponnani harbour regions were -7.41 (LRR) and -43.24 (EPR) m/year. NSM estimated erosion of -285.9 m at Ponnani harbour and 132.32 m at Ponnani beach. Mostly high to very high erosion (-2.5 to -7.4 m/year, LRR) occurred at Ponnani harbour region and Puthuponnani, and moderate erosion (-2.5 to -5 m/year, LRR) occurred at Mangalam and Purathur. The aerial view of the southern coast of the study area is shown in Fig. 8a,

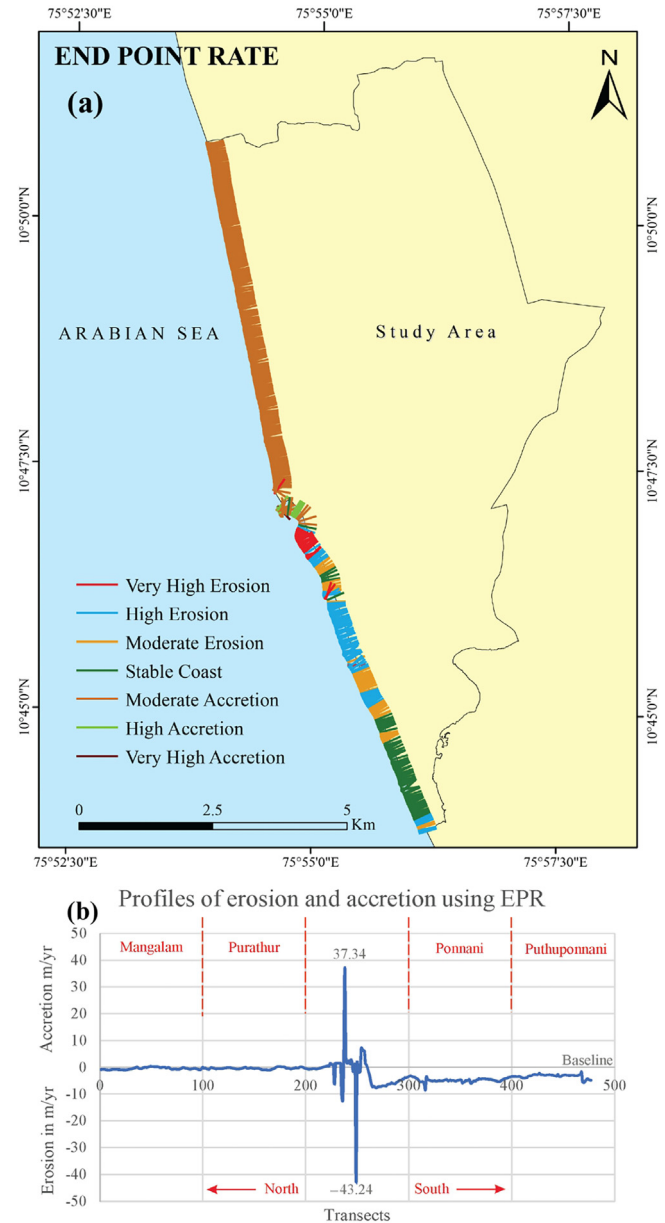


Fig. 6. Estimations of shoreline changes using EPR and profiles of erosion and accretion.

taken from the top of the Ponnani Harbour lighthouse. Fig. 8b and c shows the collapse of houses on the beach due to intense coastal erosion in the southern sector, particularly on Ponnani and Puthuponnani beaches.

5.2.5.2. Northern sector. A jetty of about 360 m in length at the meeting point of Bharathapuzha and Tirur river mouth facilitates the passage of barges and fishing boats. This sector experienced more accretion and moderate coast compared to erosion. The maximum erosion rates were -8.3 m/year in LRR at Purathur, and the maximum rates of accretion were 0.79 m/year (LRR) near the Mangalam. The NSM values showed that the shoreline retreated by -33.28 m and advanced by 8.83 m. The moderate to very high accretion (2.5 – 5 m/year, LRR) occurred at Mangalam and Purathur. The rapid accretion rate at Mangalam was induced by sediment inflow by the Bharathapuzha River in the southern region. The construction of jetties caused the erosion at Purathur. Along the jetty,

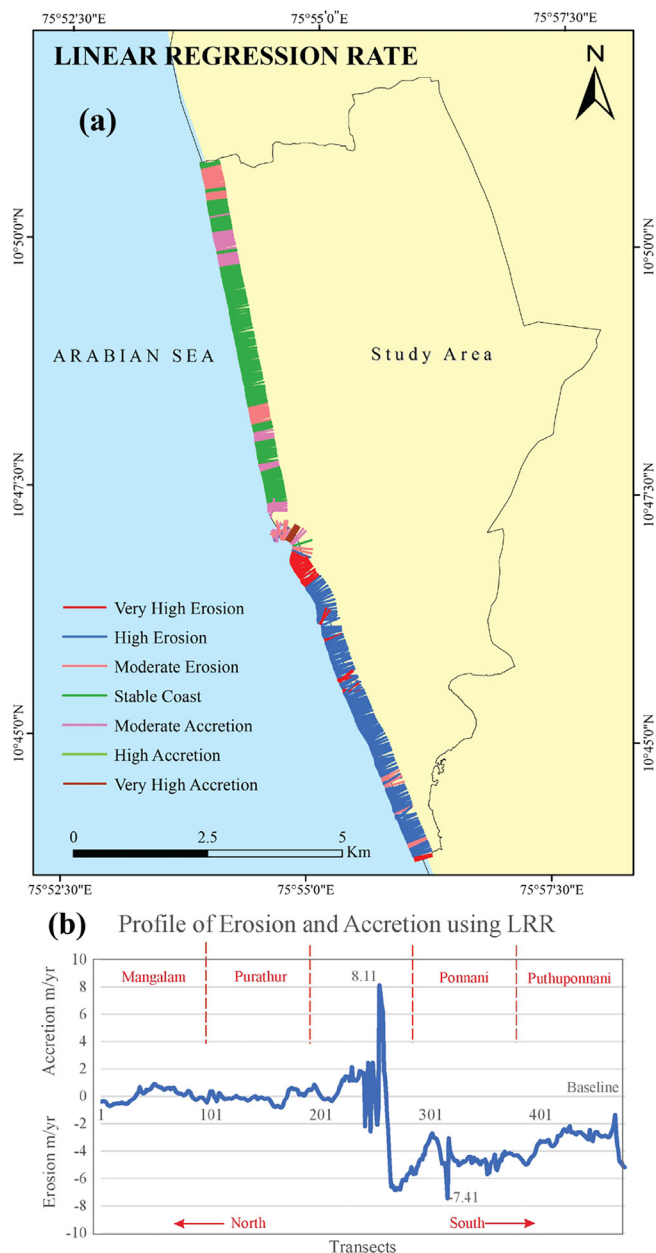


Fig. 7. Estimations of shoreline changes using LRR and profiles of erosion and accretion.

the littoral drift of sediments is holed, reducing the sediment supply.

5.3. Prediction of shoreline using Kalman filter model

The position of shorelines of the study area in the years 2030 and 2040 is predicted using the Kalman filter model (Fig. 9). Kalman filter model is a mathematical model to predict the position of the shoreline in the future by using derived values of LRR from DSAS. The model was used to forecast the position of shorelines in 2030 and 2040, compared to the shoreline position in 2020. The coastline area is divided into four zones for better understanding: Mangalam, Purathur, Ponnani, and Puthuponnani. According to the Kalman filter model, the northern sector coast will be moderate, with a slight range of accretion in Mangalam and a minor range of erosion in Purathur. The model forecasts a hazardous scenario along the southern coast for the next two decades.

The majority of the coast will be vulnerable to severe erosion, beach loss, and shoreline invasion of land. Ponnani and Puthuponnani beaches may have experienced a 100-metre shift on average. If the model predicted the shoreline shifts, various human settlements and other infrastructure setups in the southern part of the study area, including PFH, will be affected.

6. Discussion

A major shift in the shoreline at the north and south ends of the Ponnani breakwater following construction was examined by Paravath et al. in 2021. This study found that a significant erosion process occurred at the southern end, whereas the beach was reversibly accreting at the northern end. Using Google Earth, NRSC, USGS satellite sources, and field-based measurements from 1979 to 2017, they calculated LRR values for both sides. In order to examine the position of the shoreline during each phase and determine the impact of the new construction on the changes in the shoreline, they divided the time period into three categories: before, during, and after the construction of breakwaters. They concluded that by stopping the coast from being severely eroded, the breakwaters at the Ponnani River Estuary had made it possible to build a fishing vessel passage that was safe and passable. In contrast, our study uses comparable technology to identify the causes of changes in land use and shoreline changes. Additionally, we looked into statistical shoreline position measurement factors such as SCE, LRR, EPR and NSM. Using the Kalman filter model, the future position of the shoreline was also predicted.

The coastline of India is changing as a consequence of several human interventions (Ramesh et al., 2013). Over the last five decades, the areas subjected to erosion should be portrayed as high-risk zones from natural and anthropogenic phenomena (Natesan et al., 2015). The structures constructed along the coast of PFH foreshore are primarily responsible for shoreline changes. The dynamic nature of the PFH shoreline is incompatible with these developments. These factors have made it difficult to identify locations susceptible to long-term and short-term erosion, as erosion rates have frequently increased over the past ten years. When human activity is endangered by permanent or temporary holdback of a shoreline, coastal erosion becomes a risk. Seasonal variations in shorelines tend to gradually accrete in the summer when sediments are deposited by relatively low-energy waves and fiercely erode in the winter when sediments are driven offshore by high-energy storm waves. Furthermore, attempts to halt the natural coastal process with seawalls and other complex constructions shift the problem, causing similar losses in downdrift coastal areas. Moreover, some beaches, dunes, barrier beaches, salt marshes, and estuaries are threatened to disappear if there is no sediment transport since there are no longer sand sources to feed and sustain them. The surging sea waves destroyed houses and coconut trees along the coastal belt as sea erosion at Puthuponnani worsened. The upstream end of the shoreline is meant to be protected from erosion by seawalls; however, this causes further erosion. A coast that is eroding provides sediment for sediment transport. However, the sand supply from this coastline area to the sediment transport along the surrounding shorelines will get stopped. Thus, the shorelines at the end of structures are susceptible to increased erosion when the erosion is regulated at some sections by the creation of seawalls (Genz et al., 2007). According to the study by Sheeja and Vishnu (2015), the Puthuponnani stretch had a sharp drop-in erosion rate from 1999 to 2000 to 2002–2003 and a steadily increasing trend after that. It is recommended that a management strategy be considered that involves using information, methods, tools, and institutional instruments to reduce the consequences of coastal erosion. Urbanization, sand mining, and dredging are anthropogenic activities that contribute to coastal erosion.

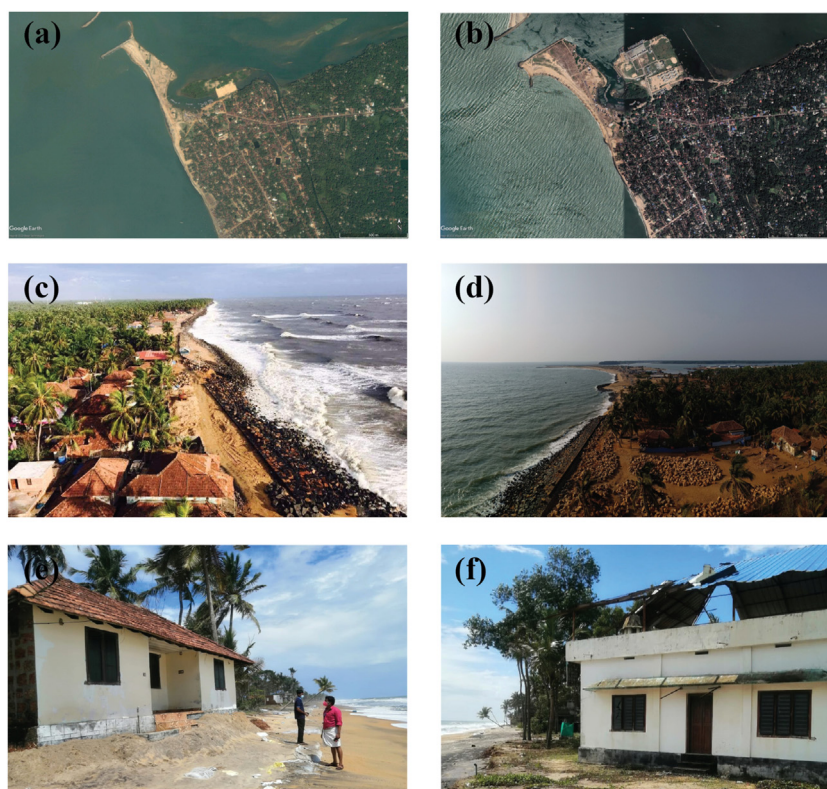


Fig. 8. (a) Aerial view of coastal areas of the southern sector, (b) Aerial view of beaches in the northern sector, (c) Collapse of houses due to coastal erosion in Ponnani beach, (d) Collapse of houses in Puthuponnani beach, (e) Satellite image of the southern sector representing the year 2004, (f) Satellite image of the southern sector representing the year 2022.

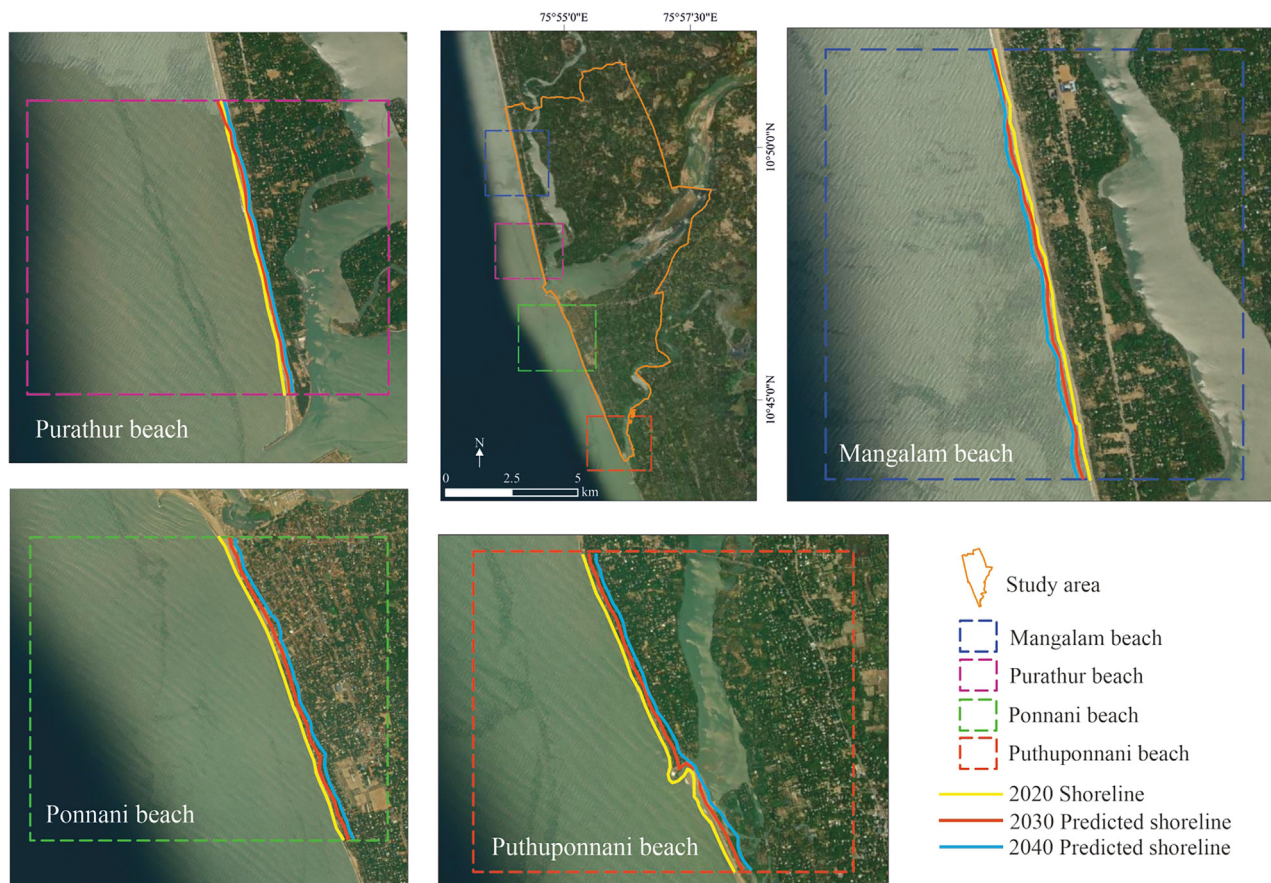


Fig. 9. Predicated shorelines at portions of northern and southern sectors in 2030 and 2040 using the changes that occurred over the last three decades.

This series of seawalls and groyne areas can be reduced by building revetments. (Linham and Nicholls, 2012; Natarajan et al., 2021). Some efficient and feasible adaptation techniques in ecologically sensitive regions include beach nourishment, sea dyke construction, wetland restoration, detached breakwaters and coastal set-backs (Linham and Nicholls, 2012; Natarajan et al., 2021).

7. Conclusions

The present study aimed to analyse the long-term shoreline changes between 1987 and 2020 by remote sensing and geospatial techniques joined with DSAS 5.0 and detected and enumerated the erosion and accretion rate of the shoreline PFH. The erosion and accretion modification were apparent along the study area, and it was influenced mainly by riverine contributions and other coastal processes. The study also focused on condensing the coastal vulnerability by predicting the possible shoreline tendency to concoct future strategies.

- The land use-land cover study showed that BSV for 2004 was 13.88%, increasing to 66.17% in 2020; likewise, the VSB decreased from 56.74% to 6.21%, indicating vegetation cover loss built-up. This means that vegetation cover will be less in 2020 compared to 2004.
- According to the most recent DSAS shoreline change analysis, erosion is more prevalent in the study area's southern sector in places like Puthuponnani and the Ponnani port area than it is in the northern sector. The shoreline retreated by approximately 7.5 m at the erosion site and advanced by approximately 8.1 m at the accretion site, each estimated using the LRR rate.
- By 2030, the shoreline at Ponnani and Puthuponnani beaches may have shifted by an average of around 100 metres; by 2040, the shoreline may have eroded another 140 metres. Given the expected outcomes, the Ponnani and Puthuponnani beach regions should be considered high erosion risk zones. The study's approach and findings could be used to manage the PFH coast sustainably.
- According to this evaluation, the PFH shoreline is eroding along its southernmost section. The main reason for erosion in PFH would be construction operations that put pressure on the coastline side. The coastline stretch's buildups have significantly increased, according to the LULC change. Precautions must be taken based on the information and data above about building any additional structures along the PFH's eroding and susceptible coastline portions.
- As a safeguard against potential hazardous situations brought on by shoreline shifts, we recommend that sophisticated scientific approaches be used along the southwest Indian coast, particularly the Kerala coast

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